

David Lawrence

Fort Collins, CO (Open to relocation)

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Portfolio: David-E-Lawrence.github.io | github.com/David-E-Lawrence

EDUCATION

Bachelor of Science in Mechanical Engineering — Aerospace Concentration | Colorado State University

Expected May 2030

- GPA: 4.0
 - Active member of university's Ram Rocketry
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EXPERIENCE

Machinist Apprentice | Oakes Manufacturing LLC | Fort Collins, CO

February 2026 – Present

- Designed SolidWorks CAD models for fabrication and assembly of parts and fixtures
 - Utilized Excel to manage tool inventory and automate tasks
 - Fabricated precision parts to $\pm 0.001''$ using CNC mill and manual lathe
 - Inspected parts using precision measurement tools to verify tolerances to $\pm 0.001''$
 - Assisted in root-cause analysis for production challenges
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PROJECTS

Rocket Flight Simulation (Python)

- Modeled rigid-body rocket dynamics using vector-based force accumulation with Euler integration
 - Implemented altitude-dependent atmosphere and aerodynamic drag including transonic drag divergence
 - Generated post-processing analysis of simulation runs including a trajectory visualization
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SKILLS

Technical:

- SolidWorks (Part Design, Assemblies, Drawings)
- Python (Simulation, Data Analysis)
- Excel (Data Tracking and Task Automation)

Manufacturing and Fabrication:

- CNC Milling (3-axis)
- Manual Lathe Operation
- Precision Measurement ($\pm 0.001''$)

Engineering:

- Design for Manufacturability (DFM)
- Technical Documentation
- Root Cause Analysis